Project Title: CONTENDER-ESTIMATION ADAPTIVE BACKOFF (CEAB): A COMPARATIVE STUDY OF WAITING STRATEGIES FOR IEEE 802.11 CSMA/CA

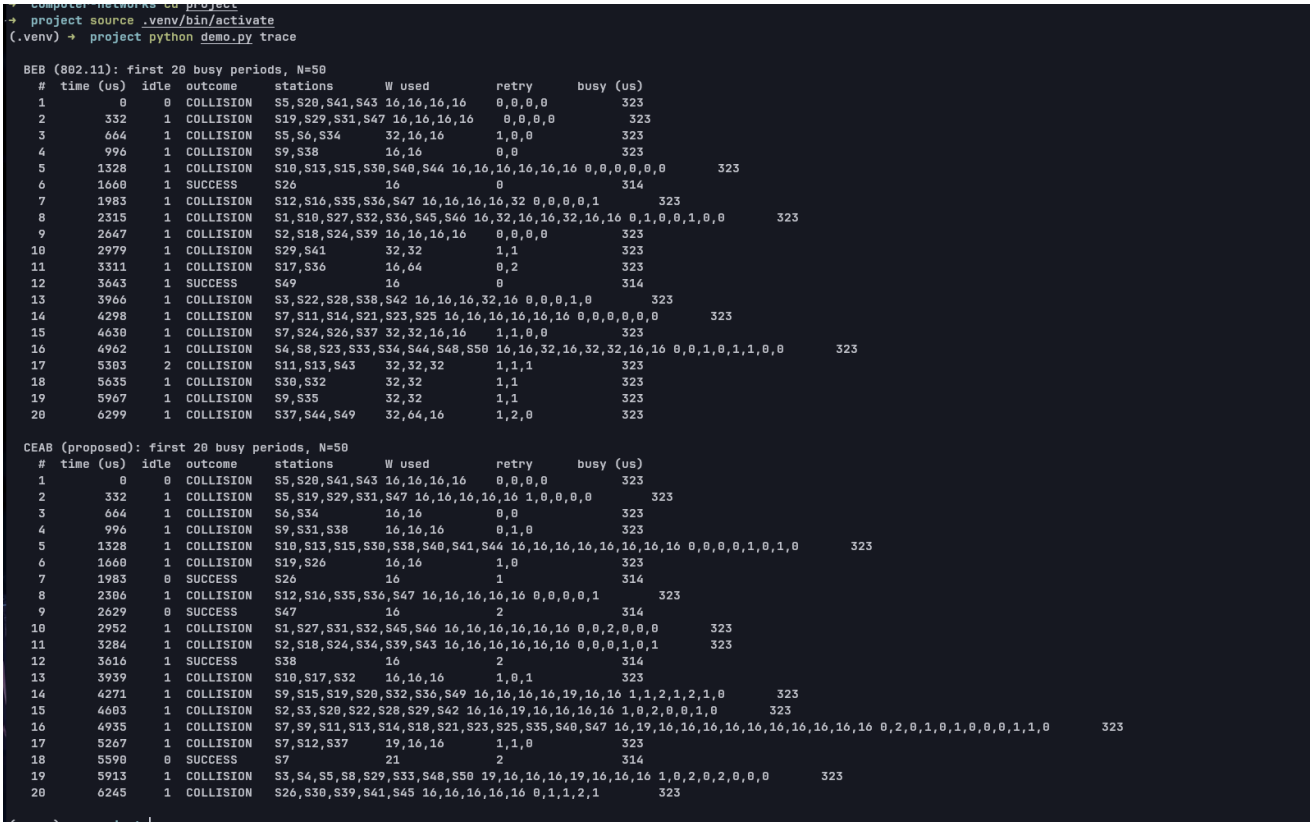
Made By: Team7 F1 slot

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GITHUB LINK: https://github.com/IAmPara0x/CEAB_CN

Implementation Screenshots:

Simulation OUTPUT:



```
→  project source .venv/bin/activate
(.venv) →  project python demo.py trace

  BEB (802.11): first 20 busy periods, N=50
    #  time (us)  idle  outcome     stations       W used         retry      busy (us)
    1          0     0  COLLISION   S5,S20,S41,S43 16,16,16,16    0,0,0,0          323
    2        332     1  COLLISION   S19,S29,S31,S47 16,16,16,16    0,0,0,0          323
    3        664     1  COLLISION   S5,S6,S34      32,16,16       1,0,0            323
    4        996     1  COLLISION   S9,S38         16,16          0,0              323
    5       1328     1  COLLISION   S10,S13,S15,S30,S40,S44 16,16,16,16,16,16 0,0,0,0,0,0       323
    6       1660     1  SUCCESS     S26            16             0                314
    7       1983     1  COLLISION   S12,S16,S35,S36,S47 16,16,16,16,32 0,0,0,0,1       323
    8       2315     1  COLLISION   S1,S10,S27,S32,S36,S45,S46 16,32,16,16,32,16,16 0,1,0,0,1,0,0       323
    9       2647     1  COLLISION   S2,S18,S24,S39 16,16,16,16    0,0,0,0          323
   10       2979     1  COLLISION   S29,S41        32,32          1,1              323
   11       3311     1  COLLISION   S17,S36        16,64          0,2              323
   12       3643     1  SUCCESS     S49            16             0                314
   13       3966     1  COLLISION   S3,S22,S28,S38,S42 16,16,16,32,16 0,0,0,1,0        323
   14       4298     1  COLLISION   S7,S11,S14,S21,S23,S25 16,16,16,16,16,16 0,0,0,0,0,0       323
   15       4630     1  COLLISION   S7,S24,S26,S37 32,32,16,16    1,1,0,0          323
   16       4962     1  COLLISION   S4,S8,S23,S33,S34,S44,S48,S50 16,16,32,16,32,32,16,16 0,0,1,0,1,1,0,0       323
   17       5303     2  COLLISION   S11,S13,S43    32,32,32       1,1,1            323
   18       5635     1  COLLISION   S30,S32        32,32          1,1              323
   19       5967     1  COLLISION   S9,S35         32,32          1,1              323
   20       6299     1  COLLISION   S37,S44,S49    32,64,16       1,2,0            323

  CEAB (proposed): first 20 busy periods, N=50
    #  time (us)  idle  outcome     stations       W used         retry      busy (us)
    1          0     0  COLLISION   S5,S20,S41,S43 16,16,16,16    0,0,0,0          323
    2        332     1  COLLISION   S5,S19,S29,S31,S47 16,16,16,16,16 1,0,0,0,0       323
    3        664     1  COLLISION   S6,S34         16,16          0,0              323
    4        996     1  COLLISION   S9,S31,S38     16,16,16       0,1,0            323
    5       1328     1  COLLISION   S10,S13,S15,S30,S38,S40,S41,S44 16,16,16,16,16,16,16,16 0,0,0,0,1,0,1,0       323
    6       1660     1  COLLISION   S19,S26        16,16          1,0              323
    7       1983     0  SUCCESS     S26            16             1                314
    8       2306     1  COLLISION   S12,S16,S35,S36,S47 16,16,16,16,16 0,0,0,0,1       323
    9       2629     0  SUCCESS     S47            16             2                314
   10       2952     1  COLLISION   S1,S27,S31,S32,S45,S46 16,16,16,16,16,16 0,0,2,0,0,0       323
   11       3284     1  COLLISION   S2,S18,S24,S34,S39,S43 16,16,16,16,16,16 0,0,0,1,0,1       323
   12       3616     1  SUCCESS     S38            16             2                314
   13       3939     1  COLLISION   S10,S17,S32    16,16,16       1,0,1            323
   14       4271     1  COLLISION   S9,S15,S19,S20,S32,S36,S49 16,16,16,16,19,16,16 1,1,2,1,2,1,0       323
   15       4603     1  COLLISION   S2,S3,S20,S22,S28,S29,S42 16,16,19,16,16,16,16 1,0,2,0,0,1,0       323
   16       4935     1  COLLISION   S7,S9,S11,S13,S14,S18,S21,S23,S25,S35,S40,S47 16,19,16,16,16,16,16,16,16,16,16,16 0,2,0,1,0,1,0,0,0,1,1,0       323
   17       5267     1  COLLISION   S7,S12,S37     19,16,16       1,1,0            323
   18       5590     0  SUCCESS     S7             21             2                314
   19       5913     1  COLLISION   S3,S4,S5,S8,S29,S33,S48,S50 19,16,16,16,19,16,16,16 1,0,2,0,2,0,0,0       323
   20       6245     1  COLLISION   S26,S30,S39,S41,S45 16,16,16,16,16 0,1,1,2,1        323
```

Simulation Graphs:

BEB VS CEAB



BEB VS CEAB VS Linear VS Quadratic

